

Part 5: Python for Data Analysis

1. Python Fundamentals

1. Introduction to Python & Setting Up the Environment
2. Variables and Data Types
3. Operators (Arithmetic, Comparison, Logical)
4. Strings and String Methods
5. Lists, Tuples, and Dictionaries
6. Conditional Statements (if/elif/else)
7. Loops (for, while)
8. Functions and Arguments
9. Error Handling (try/except)
10. Introduction to Object-Oriented Concepts (Classes — basic awareness)

2. Working with Data in Python

11. Introduction to Libraries — pip and Virtual Environments
12. NumPy Fundamentals
 - Arrays and Array Operations
 - Broadcasting
13. Introduction to Pandas
 - Series and DataFrames
 - Reading Data (CSV, Excel, SQL, JSON)

3. Data Cleaning and Preparation with Pandas

14. Inspecting Data (.head(), .info(), .describe())
15. Handling Missing Values
16. Removing Duplicates
17. Data Type Conversion
18. String Cleaning and Text Processing
19. Filtering and Sorting Data
20. Renaming and Reordering Columns
21. Handling Outliers

4. Data Transformation

22. Selecting and Indexing (.loc, .iloc)
23. Creating and Modifying Columns
24. GroupBy and Aggregation
25. Merging, Joining, and Concatenating DataFrames
26. Pivot Tables in Pandas
27. Working with Dates and Time Series Data
28. Applying Functions (.apply(), lambda functions)

5. Exploratory Data Analysis (EDA) in Python

29. Descriptive Statistics with Pandas
30. Correlation Analysis
31. Identifying Patterns and Trends
32. Working with Categorical Data

6. Data Visualization in Python

- 33. Introduction to Matplotlib
- 34. Introduction to Seaborn
- 35. Common Chart Types (bar, line, histogram, scatter, box plot)
- 36. Customizing and Styling Charts
- 37. Introduction to Interactive Visualization (Plotly — overview)

7. Introduction to Statistics with Python

- 38. Probability Distributions (overview)
- 39. Hypothesis Testing (overview)
- 40. Using SciPy for Statistical Tests

8. Working with Databases and APIs

- 41. Connecting Python to SQL Databases
- 42. Running SQL Queries from Python
- 43. Introduction to APIs and Pulling Data with requests
- 44. Working with JSON Data

9. Automating and Reporting

- 45. Writing Reusable Scripts
- 46. Automating Repetitive Data Tasks
- 47. Exporting Cleaned Data and Reports (to Excel, CSV, PDF)

10. Version Control and Portfolio Readiness

- 48. Using Git and GitHub with Python Projects
- 49. Structuring a Data Analysis Project (folders, notebooks, README)
- 50. Publishing a Python Project to GitHub